

By being able to diagnose deviations and anomalies in a network in real time and employing supercomputing, any fixes can be quickly done. When put to monitor and detect anomalies in the network, AI and machine learning (ML) can help in predicting and identifying vulnerabilities and risks by correlating centrally available data to draw statistical inferences from it.

The same AI that could lead to a ransomware attack through personalisation of data to fool a person, system, or even develop a malware that adapts to new vulnerabilities would instead make it possible to protect against ransomware, network intrusions, distributed denial-of-service attacks, and increasingly sophisticated malware, among other cyber-attacks. Code patterns used in previous attacks provide valuable insights for the detection of new threats.

Authentication methods like fingerprint readers or facial recognition systems are all an attempt to provide better protection than

just using passwords. Using digital signatures and specialised hardware to store and safeguard passwords are some other ways of ensuring the same. Even hardware like USB fobs can act as a digital key for accessing assets. Emerging technologies will be capable of doing the advanced implementation of what humans have done but at better speed, scale, and scope.

Investments in research and development in cybersecurity are needed in both the public and private sectors. Organisations need to keep up with the pace of emerging threats. There is a global shortage of people in cyber skills. Cybersecurity company Endgame created an open source data set to assist researchers and practitioners in training and testing of new algorithms a few years ago.

It is necessary to make employees aware of the risks too so that they do not aid cyberattacks. This is quite evident from the recent hacking of prominent Twitter accounts. As per the reports, the accused had convinced

the social networking company's employees that he worked at their IT department and was a co-worker. According to a 2018 Gartner survey, nearly 88 per cent of 3,000 chief information officers (CIOs) said that cybersecurity was a top focus.

Several cybersecurity firms such as Veridu, Avaleris, WISEKey, Signifyd, Socure, and ThreatMetrix have been working on designing complete end-to-end protection solutions using the latest technologies like AI. Governments worldwide need to develop strict regulations to ensure that companies follow proper measures and laws that penalise perpetrators for their involvement in cyberattacks.

According to a report by Grand View Research Inc., the global AI market is estimated to reach 733.7 billion dollars in 2027. **END** 

By: Ayushee Sharma

The author was a technology journalist at EFY until recently.

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The product is offered for businesses and individuals. It allows users to learn English with the help of chatbots or tutors. The platform was launched in August 2017 by Ninad Vengurlekar and Amit Bhadbhade.

Speedlabs: This is an online learning and practicing platform. Speedlabs uses AI and ML technologies to provide customised and highly engaging practice platform and learning resources. It was founded by IIT and IIM alumni.

Is AI replacing teachers?

With so many new AI-based technologies around and a plethora of learning platforms available online, students have so many flexible options

available for them. But the question is, what about teachers? Is AI a blessing for teachers? What benefits do teachers get from AI technology? Some people may even ask—do we really need traditional teachers in classrooms in future?

It's true that AI technology brings a lot of benefits to various fields, including education. Many researchers claim that AI and ML can increase the level of education. The latest innovations allow developers to teach a computer to do complicated tasks. It lends the opportunity to improve learning processes. Researchers have been able to make

robots and AI-based solutions to teach humans in a limited way. But replacing a human teacher or professor by an AI-robot or AI-based solution is impossible. AI provides many benefits not only to students but teachers as well.

AI technology assists teachers to perform better, improves the teaching process, and reduces the burden of teachers, which, in turn, improves the students' learning curves and experiences. **END** 

By: Sani Theo

The author is a technical editor at EFY.

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